



Applied Machine Learning for Text Analysis (24ALT3101)

MathSLM – A TRANSFORMER-BASED SMALL LANGUAGE MODEL FOR MATHEMATICAL REASONING

Group - 17

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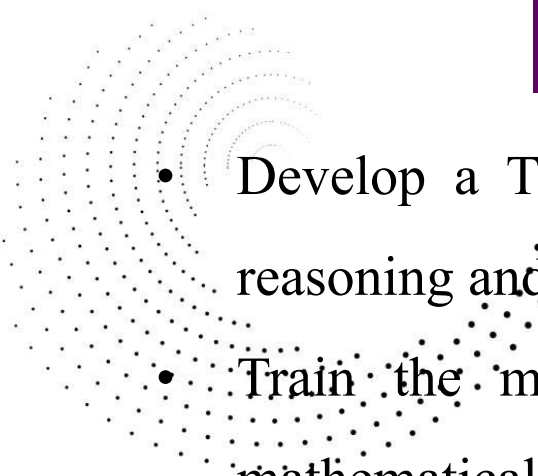
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PROBLEM STATEMENT

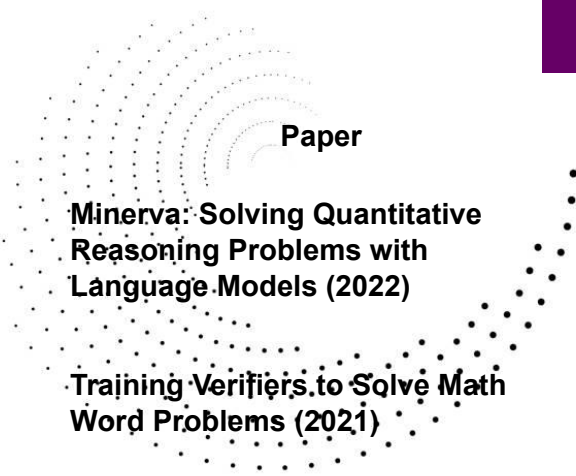
Existing Large Language Models achieve strong mathematical reasoning but require significant computational resources, making them impractical for resource-constrained environments, while traditional rule-based and machine learning approaches lack generalization across diverse mathematical problems. Therefore, there is a need for a lightweight Transformer-based Small Language Model capable of efficient step-by-step mathematical reasoning.



OBJECTIVES

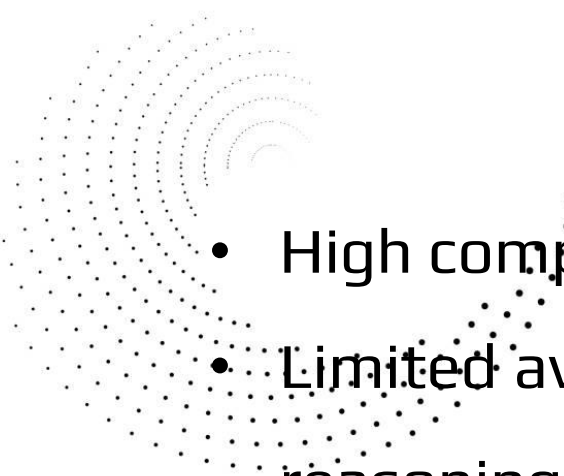
- Develop a Transformer-based Small Language Model (MathSLM) for mathematical reasoning and problem solving.
- Train the model on publicly available mathematical reasoning datasets to learn mathematical concepts and solution patterns.
- Generate accurate step-by-step solutions for mathematical problems using autoregressive text generation.
- Evaluate the model using metrics such as validation loss, perplexity, token prediction accuracy, and mathematical answer accuracy.
- Deploy the trained model through an interactive Streamlit-based web application for real-time mathematical query solving.
- Provide a lightweight and computationally efficient alternative to large language models for educational and resource-constrained environments.

LITERATURE SURVEY



Paper	Model / Approach	Dataset	Key Finding	Research Gap
Minerva: Solving Quantitative Reasoning Problems with Language Models (2022)	PaLM Fine-tuned for Mathematics	MATH, GSM8K, MMLU-STEM	Demonstrated strong mathematical reasoning using large language models with chain-of-thought generation.	Requires extremely large models and significant computational resources.
Training Verifiers to Solve Math Word Problems (2021)	GPT-3 + Verifier Model	GSM8K	Verification models significantly improve mathematical reasoning accuracy over direct generation.	Computationally expensive and dependent on large-scale models.
TinyLlama: An Open-Source Small Language Model (2024)	Decoder-Only Transformer	SlimPajama	Small language models can achieve competitive performance while requiring substantially fewer computational resources.	Not specialized for mathematical reasoning tasks.
Phi-2: The Surprising Power of Small Language Models (2023)	Transformer-based Small Language Model	Synthetic Textbooks + Web Data	High-quality training data enables compact language models to perform strong reasoning tasks.	General-purpose model without domain-specific mathematical optimization.
Qwen2.5 Technical Report (2024)	Decoder-Only Transformer	Large Multi-domain Corpus	Demonstrates strong reasoning and multilingual capabilities across various benchmark tasks.	Higher computational requirements and not designed specifically for lightweight mathematical applications.

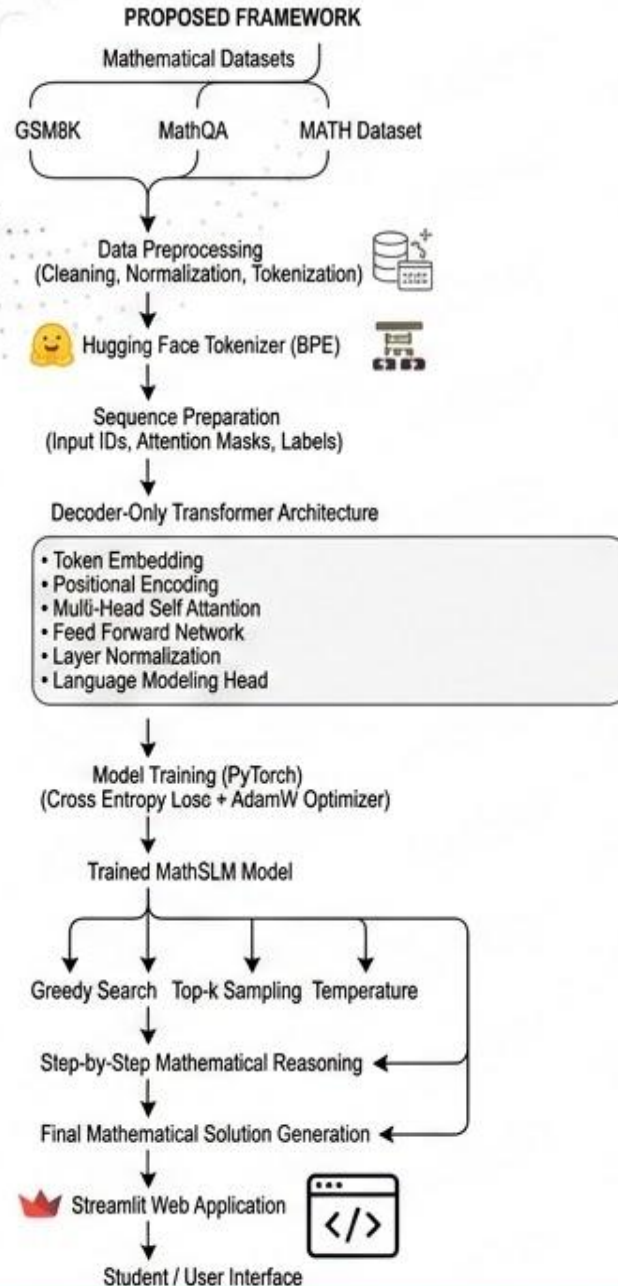
Observation
Current mathematical reasoning models achieve high accuracy but require large computational resources, creating the need for a lightweight Transformer-based Small Language Model.



RESEARCH GAP

- High computational and memory requirements for deployment.
- Limited availability of lightweight models specialized for mathematical reasoning.
- Existing small language models are primarily general-purpose and not optimized for step-by-step mathematical problem solving.
- Traditional rule-based systems struggle to generalize across diverse mathematical domains.
- Lack of efficient Transformer-based Small Language Models for educational and real-time mathematical reasoning applications.

PROPOSED FRAMEWORK



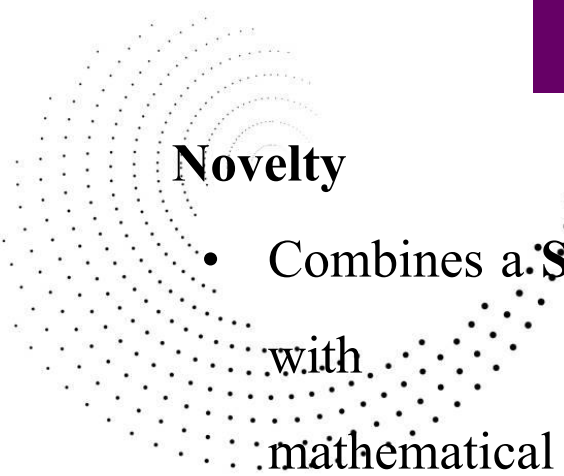
- High computations for deployment.
- lightweight model specialized for mathematical reasoning.
- Existing are primarily general-purpose and not optimized for problem solving.
- Traditional rule to generalize across diverse mathematical domains.
- Lack of n-based Small Language Models for mathematical reasoning applications.



INNOVATION

- Lightweight Transformer-based Small Language Model specialized for mathematical reasoning.
- Domain-specific training using mathematical reasoning datasets (GSM8K, MathQA, MATH).
- Step-by-step mathematical solution generation through autoregressive decoding.
- Computationally efficient alternative to large language models for educational applications.
- Interactive web-based mathematical problem-solving interface using Streamlit.
- End-to-end implementation of a Transformer-based mathematical reasoning pipeline using PyTorch.

CREATIVITY & NOVELTY



Novelty

- Combines a **Small Language Model** with Transformer-based mathematical reasoning.
- Designed specifically for mathematical problem solving.
- Targets resource-constrained educational environments.
- Provides a foundation for domain-specific mathematical intelligence.

Creativity

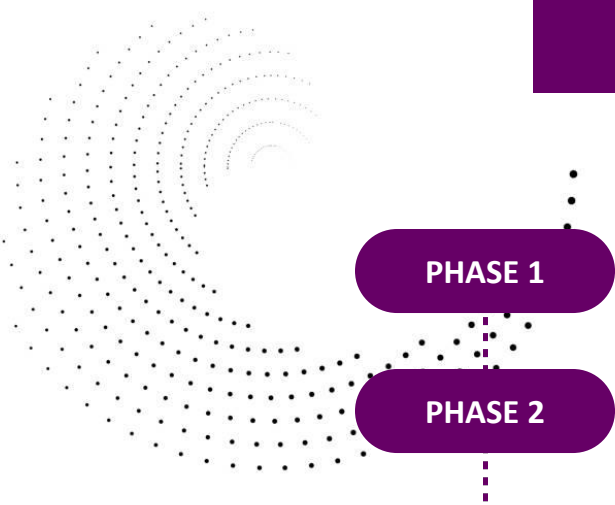
- Generates intermediate reasoning steps instead of only final answers.
- Uses multiple configurable generation strategies.
- Provides an interactive interface for real-time mathematical queries.
- Focuses on computationally efficient mathematical reasoning.

FEASIBILITY

Technologies

1. **Deep Learning Framework** - PyTorch
2. **Model Architecture** - Decoder-only Transformer
3. **Training** - Supervised training, Next-token prediction, Autoregressive text generation
4. **Datasets** - GSM8K , MathQA , MATH Dataset , NuminaMath (*Optional*)

PROJECT PLAN



PHASE 1

Literature Survey & Requirement Analysis

PHASE 2

Mathematical Dataset Collection & Data Preprocessing

PHASE 3

Tokenization & Dataset Preparation

PHASE 4

Transformer Model Development & Training

PHASE 5

Model Evaluation & Mathematical Reasoning Testing

PHASE 6

Streamlit Web Application Development & Model Integration

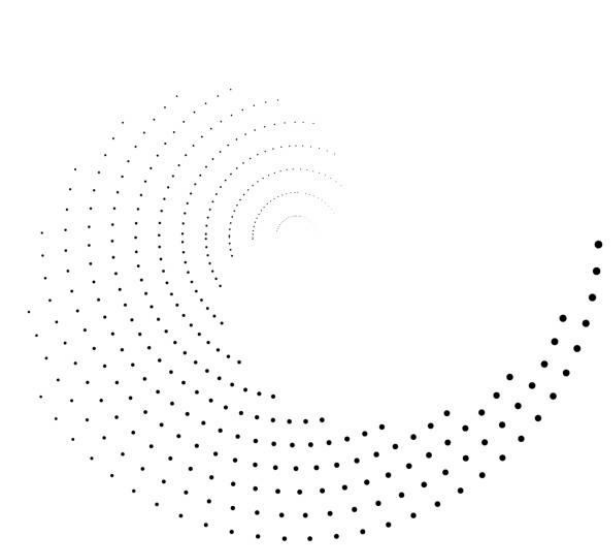
PHASE 7

Documentation, Testing & Final Presentation



EXPECTED OUTCOME

- Efficient Small Language Model for mathematical reasoning.
- Step-by-step mathematical solution generation.
- Interactive mathematical reasoning through a web interface.
- Improved learning experience
- Enhanced conceptual understanding



THANK YOU